

Project Report

Project Title : Hospital Operations & Patient Analysis Dashboard

Tool Used : Microsoft Power BI Desktop

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2. Introduction / Project Overview

This project involves building an interactive dashboard using **Power BI** to analyze hospital patient data. The dataset contains information about patients, treatments, costs, length of stay, outcomes, and satisfaction levels.

Background

Hospitals generate large volumes of data related to patient care, treatment procedures, costs, and outcomes. Analyzing this data helps improve hospital efficiency, reduce readmissions, and enhance patient satisfaction.

Purpose of Analysis

To design a visual and interactive dashboard that helps stakeholders (hospital management, doctors, administrators) understand patient trends, financial performance, and risk factors.

Business Problem

Hospitals often face challenges such as:

- High readmission rates
- Increasing treatment costs
- Identifying high-risk conditions
- Improving patient satisfaction

This dashboard provides insights to address these issues.

3. Objectives of the Project

- Analyze total patients and hospital revenue
- Identify high-cost conditions and procedures
- Evaluate patient readmission and recovery rates
- Analyze patient satisfaction levels
- Understand relationship between cost and length of stay
- Provide data-driven insights for hospital improvement

4. Dataset Description

Dataset Name: Hospital Data Analysis

Source: Kaggle dataset (CSV file)

Number of Rows & Columns: ~1000 rows, multiple columns

Key Fields:

Column Name	Description
Patient_ID	Unique patient identifier
Age	Patient age
Gender	Male / Female
Condition	Disease/health condition
Procedure	Treatment performed
Cost	Treatment cost
Length_of_Stay	Number of days admitted
Readmission	Whether patient re-admitted
Outcome	Recovered / Stable
Satisfaction	Patient rating

5. Data Cleaning / Preparation

- Imported dataset into Power BI
- Checked for null values and handled appropriately
- Ensured correct data types (numeric, categorical)
- Created calculated measures using DAX
- No major duplicates found
- Data structured for analysis

6. Data Model

Single table dataset used.

No relationships or joins required.

7. Calculated Fields (DAX)

Total Patients

Total Patients = COUNT('hospital data analysis'[Patient_ID])

Total Revenue

Total Revenue = SUM('hospital data analysis'[Cost])

Average Treatment Cost

Avg Treatment Cost = AVERAGE('hospital data analysis'[Cost])

Readmission Rate %

Readmission Rate % =

DIVIDE([Readmission Count], [Total Patients]) * 100

Recovery Rate %

Recovery Rate % =

DIVIDE([Recovered Count], [Total Patients]) * 100

Average Cost per Day

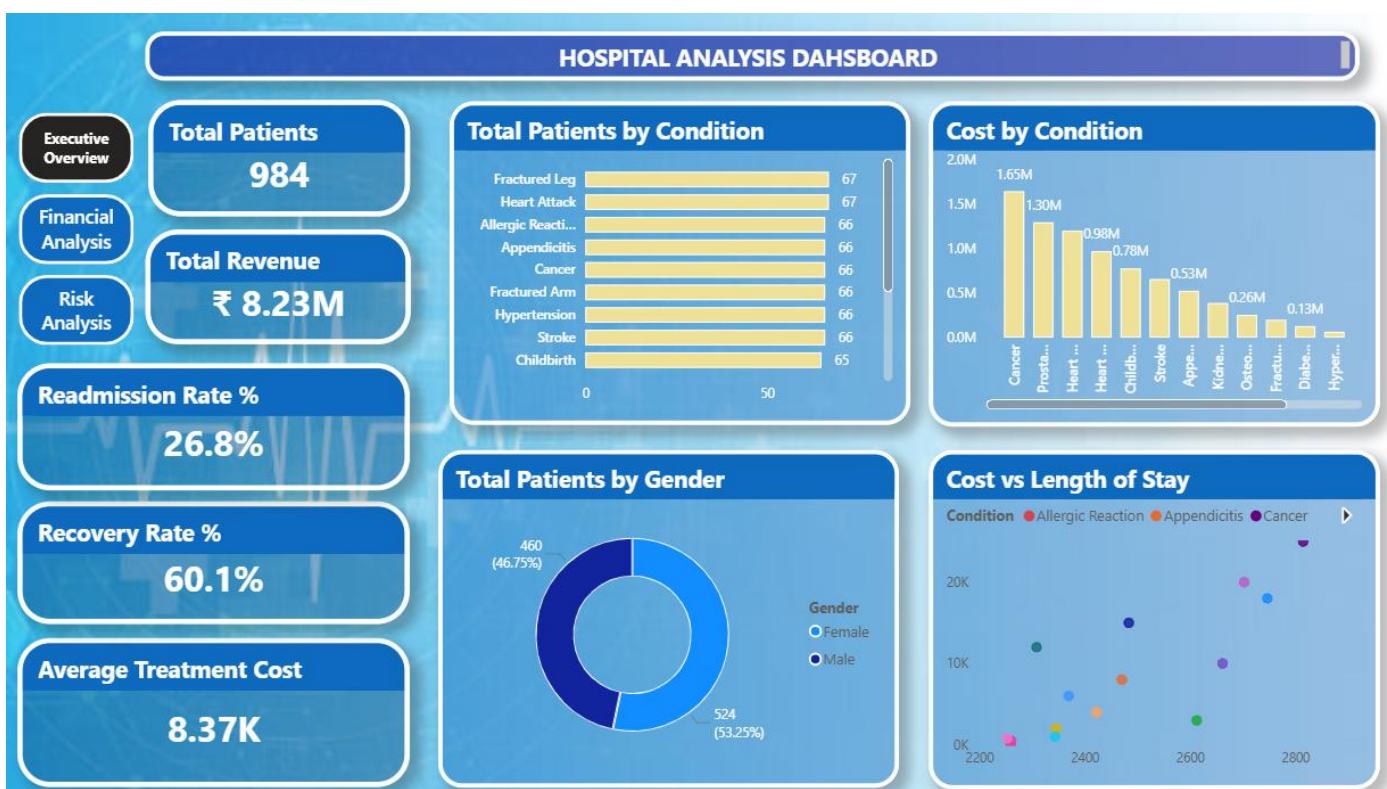
Avg Cost per Day =

DIVIDE(SUM([Cost]), SUM([Length_of_Stay]))

8. Dashboard Design

The dashboard consists of **3** pages:

Page 1 – Executive Overview



KPI Cards:

- **Total Patients**

This KPI card displays the total number of patients recorded in the dataset. It provides a quick overview of hospital patient volume. It helps in understanding overall hospital load.

- **Total Revenue**

This KPI shows the total treatment cost collected from all patients. It represents the hospital's overall revenue. It helps evaluate financial performance.

- **Readmission Rate %**

This KPI indicates the percentage of patients who were re-admitted after treatment. It helps measure hospital quality and patient recovery effectiveness. Higher values indicate potential issues.

- **Recovery Rate %**

This KPI shows the percentage of patients who successfully recovered. It reflects treatment success rate. A higher recovery rate indicates better healthcare performance.

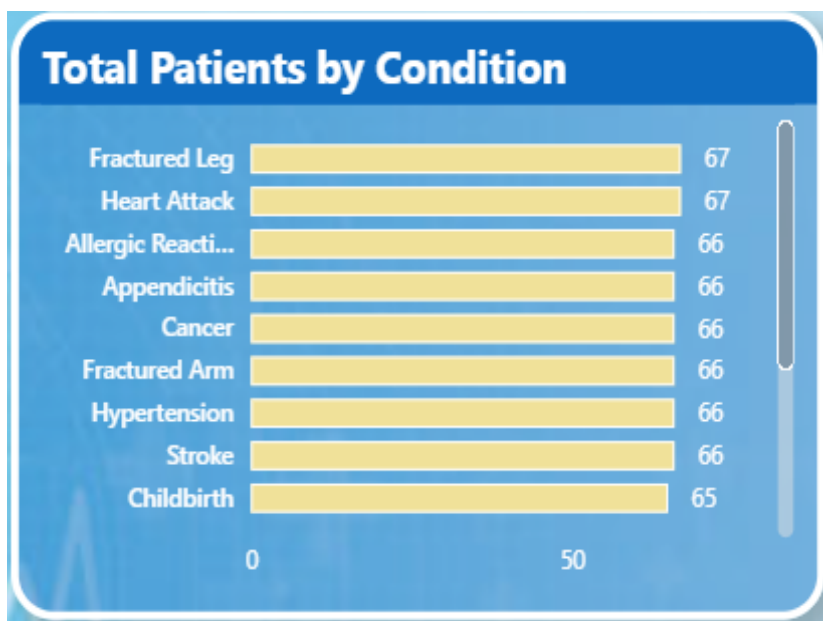
- **Average Treatment Cost**

This KPI displays the average cost incurred per patient. It helps understand treatment expense levels. Useful for cost optimization analysis.

Visuals:

- **Patients by Condition (Bar Chart)**

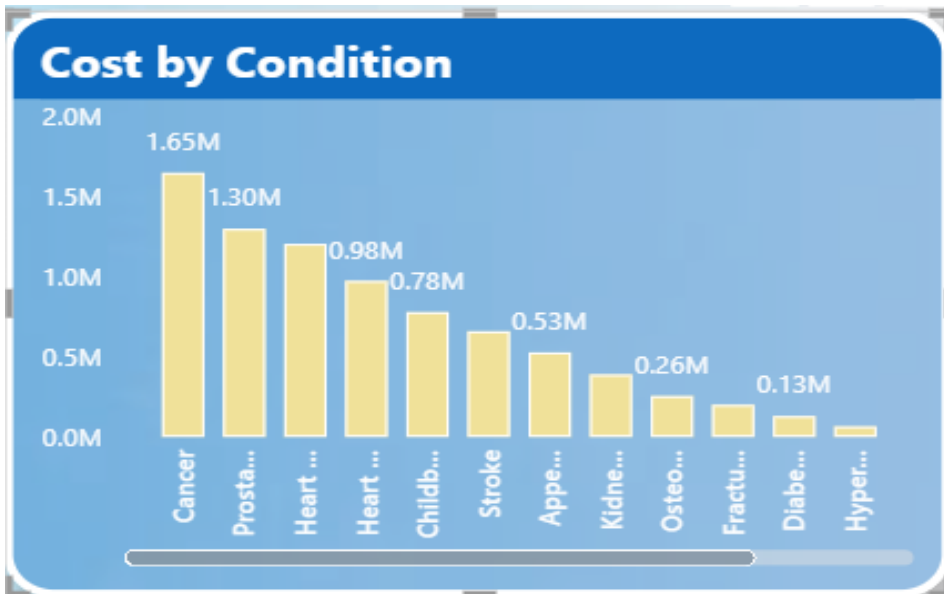
This bar chart shows the number of patients for each medical condition. It helps identify the most common diseases. Useful for resource planning.



Condition	Total Patients
Fractured Leg	67
Heart Attack	67
Allergic Reaction	66
Appendicitis	66
Cancer	66
Fractured Arm	66
Hypertension	66
Stroke	66
Childbirth	65
Diabetes	65
Heart Disease	65
Kidney Stones	65
Prostate Cancer	65
Respiratory Infection	65
Osteoarthritis	64

- **Cost by Condition (Column Chart)**

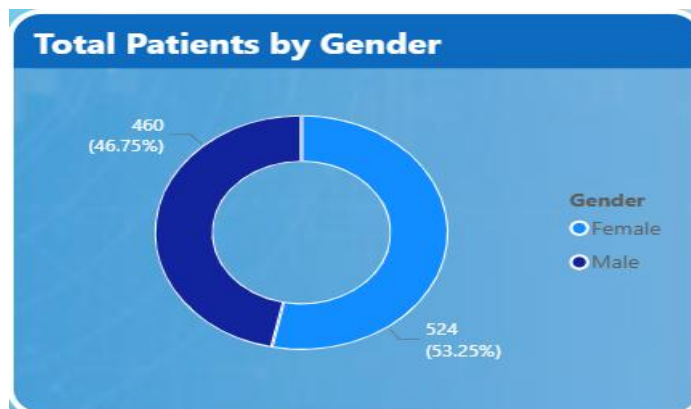
This column chart represents total treatment cost for each condition. It helps identify high-cost diseases. Useful for financial analysis.



Condition	Cost
Cancer	1650000
Prostate Cancer	1300000
Heart Attack	1206000
Heart Disease	975000
Childbirth	780000
Stroke	660000
Appendicitis	528000
Kidney Stones	390000
Osteoarthritis	256000
Fractured Leg	201000
Diabetes	130000
Hypertension	66000
Respiratory Infection	52000
Fractured Arm	33000
Allergic Reaction	6600

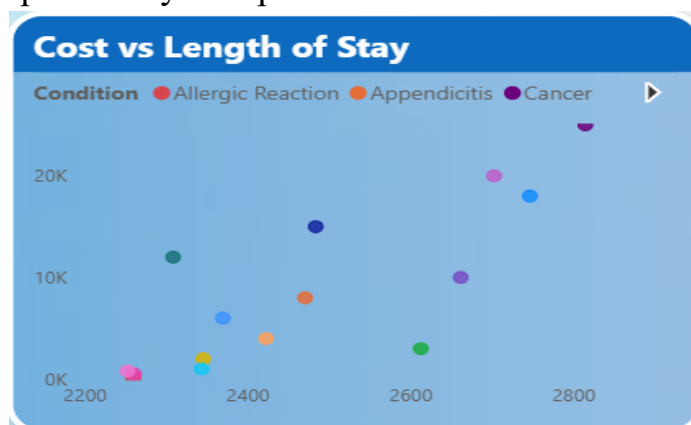
- **Gender Distribution (Donut Chart)**

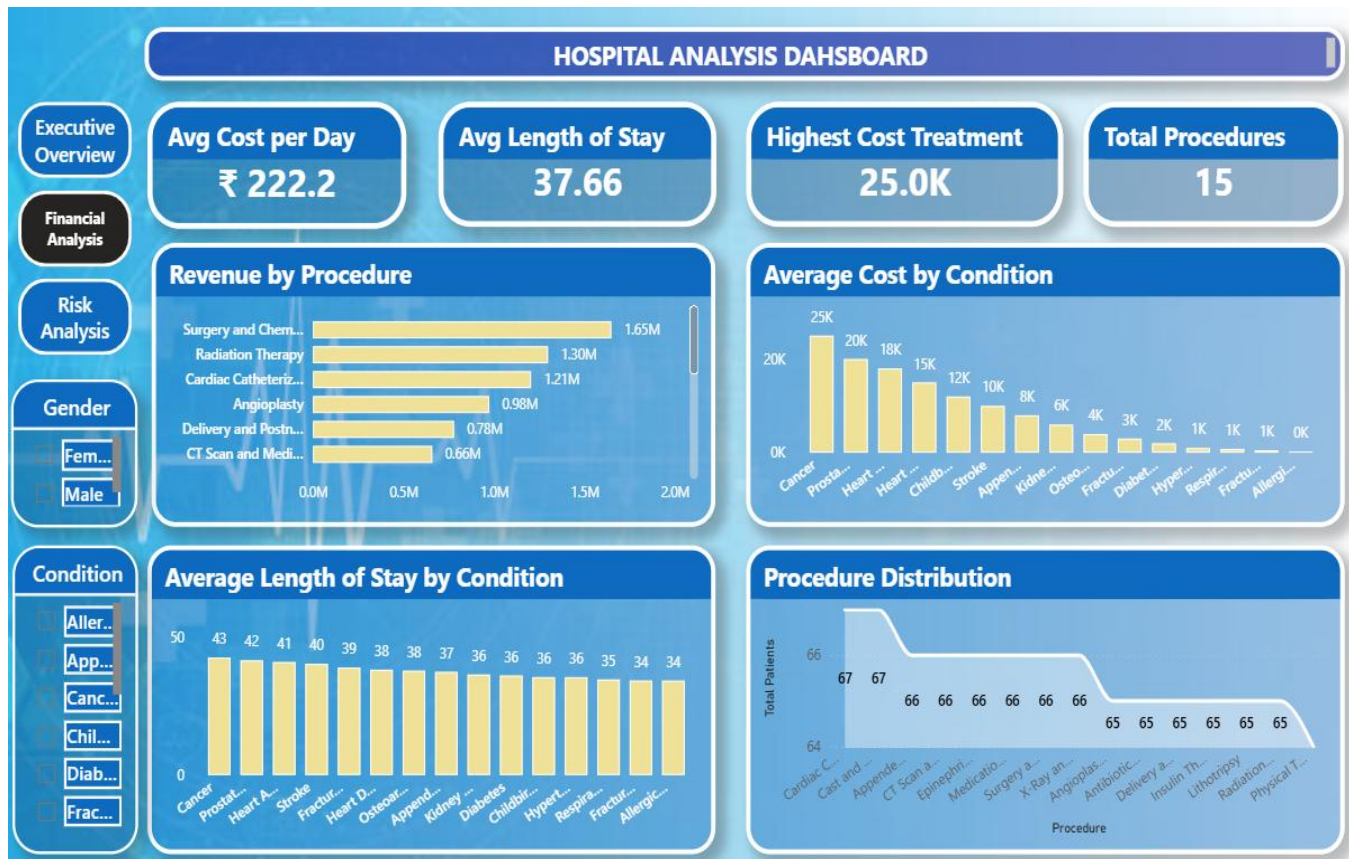
This donut chart shows the distribution of patients based on gender. It helps understand demographic patterns. Useful for population analysis.



- **Cost vs Length of Stay (Scatter Plot)**

This scatter plot shows the relationship between hospital stay duration and treatment cost. It helps identify cost patterns and outliers. Useful for efficiency analysis.





KPI Cards:

- Avg Cost per Day**

This KPI shows the average cost incurred per day of hospital stay. It helps evaluate daily treatment expenses. Useful for cost control.

- Avg Length of Stay**

This KPI displays the average number of days patients stay in the hospital. It helps measure hospital efficiency. Longer stays may increase costs.

- Highest Cost Treatment**

This KPI identifies the maximum treatment cost recorded. It highlights the most expensive case. Useful for analyzing extreme values.

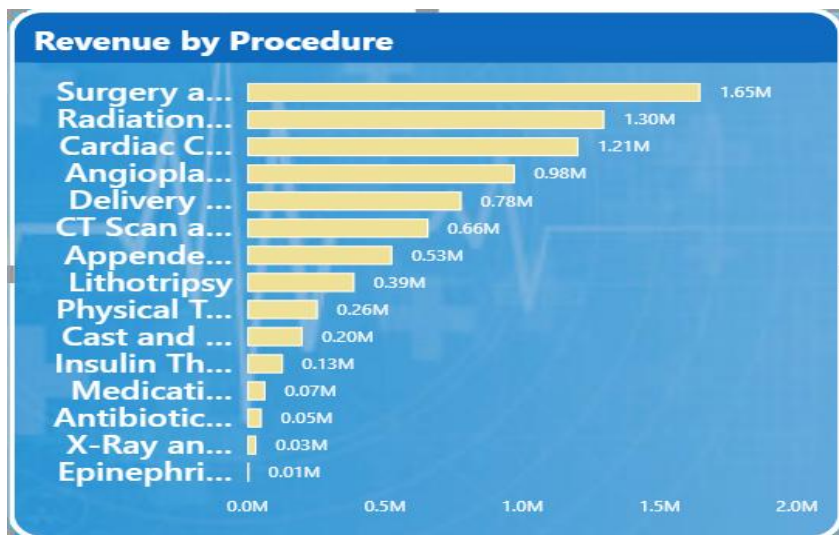
- Revenue per Patient**

This KPI shows the number of distinct procedures performed. It indicates the variety of treatments provided. Useful for operational analysis.

Visuals:

- Revenue by Procedure**

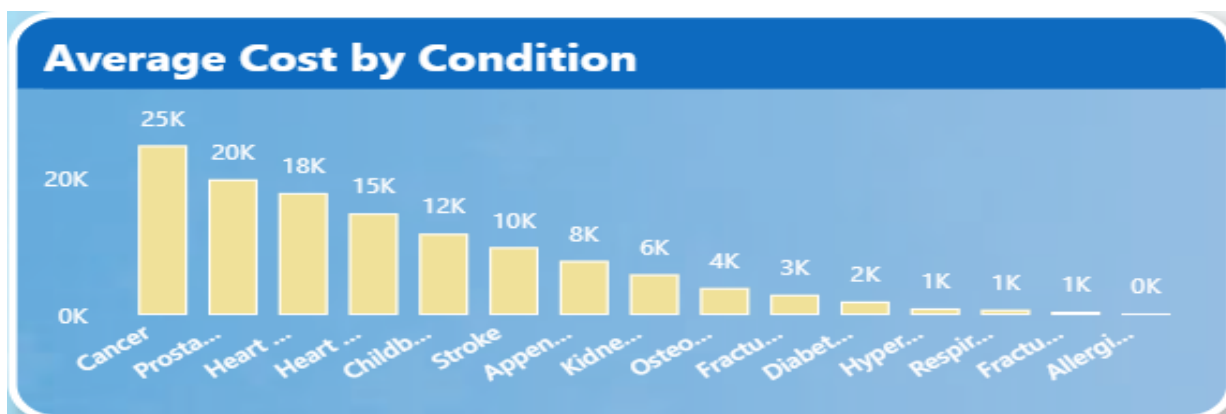
This bar chart shows total revenue generated by each procedure. It helps identify the most profitable treatments. Useful for financial planning.



Procedure	Sum of Cost
Surgery and Chemotherapy	1650000
Radiation Therapy	1300000
Cardiac Catheterization	1206000
Angioplasty	975000
Delivery and Postnatal Care	780000
CT Scan and Medication	660000
Appendectomy	528000
Lithotripsy	390000
Physical Therapy and Pain Management	256000
Cast and Physical Therapy	201000
Insulin Therapy	130000

- Average Cost by Condition**

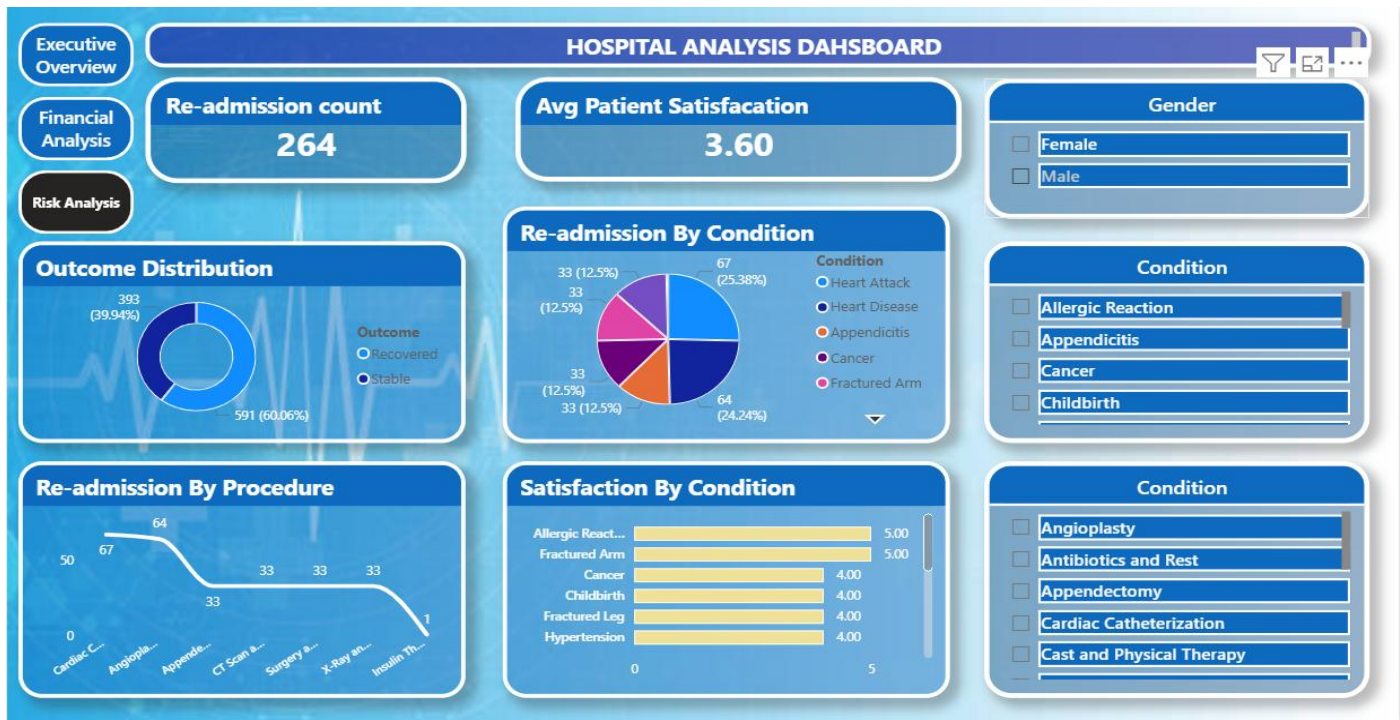
This chart displays the average cost for each condition. It helps identify expensive treatments. Useful for budgeting decisions.



- Length of Stay by Condition**

This chart shows average hospital stay for each condition. It helps identify conditions requiring longer treatment. Useful for planning hospital resources.





KPI Cards:

- Readmission Count**

This KPI shows the number of patients who were readmitted. It helps identify high-risk cases. Useful for improving treatment quality.

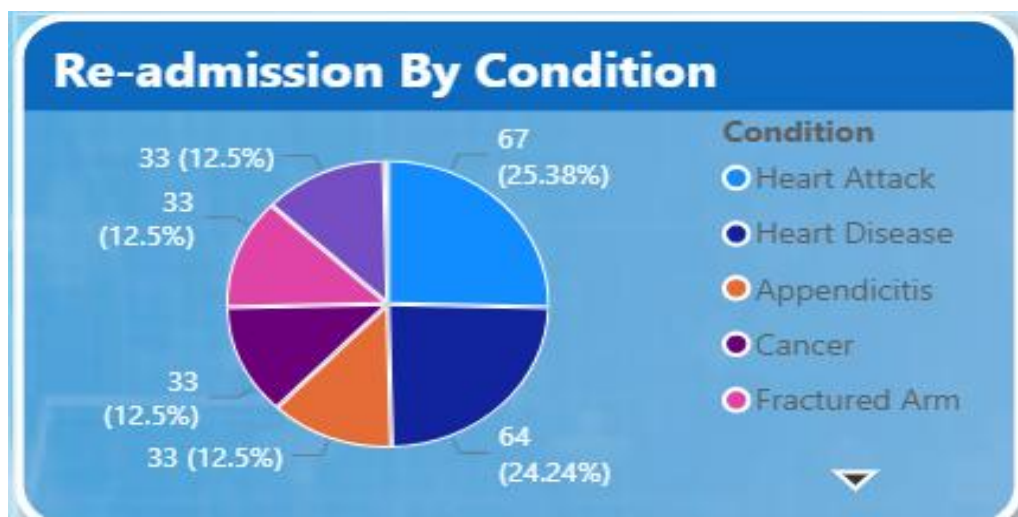
- Avg Patient Satisfaction**

This KPI displays the average satisfaction score given by patients. It reflects hospital service quality. Higher values indicate better patient experience.

Visuals:

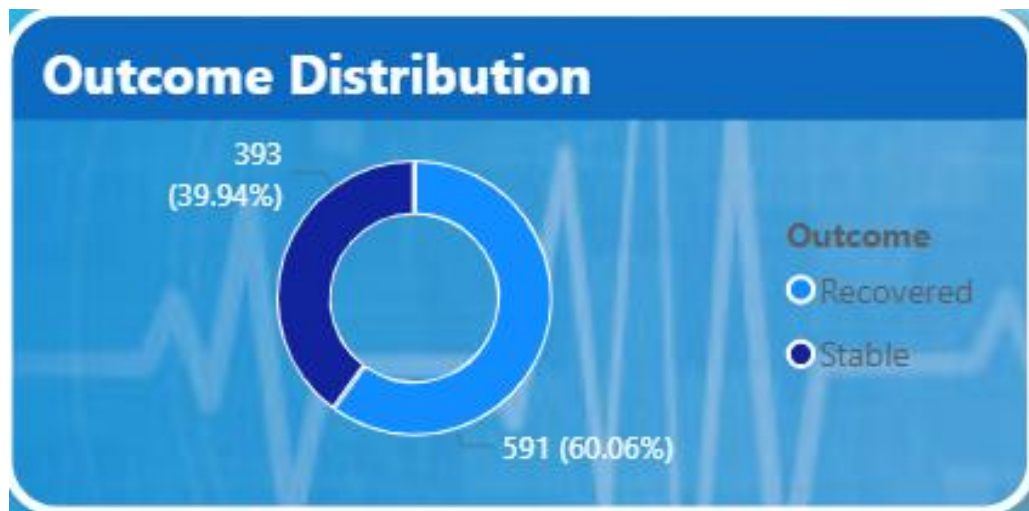
- Readmission by Condition**

This chart shows which conditions have higher readmission cases. It helps identify risk-prone diseases. Useful for preventive measures.



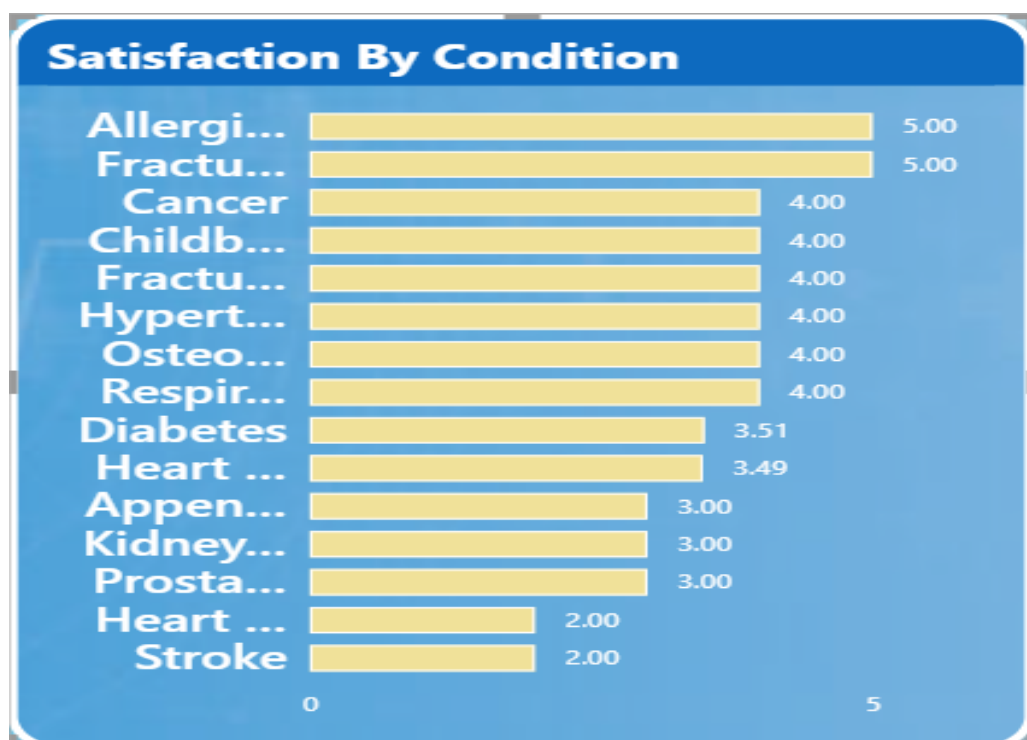
- **Outcome Distribution**

This donut chart shows the distribution of patient outcomes such as recovered or stable. It helps evaluate treatment success. Useful for performance analysis.



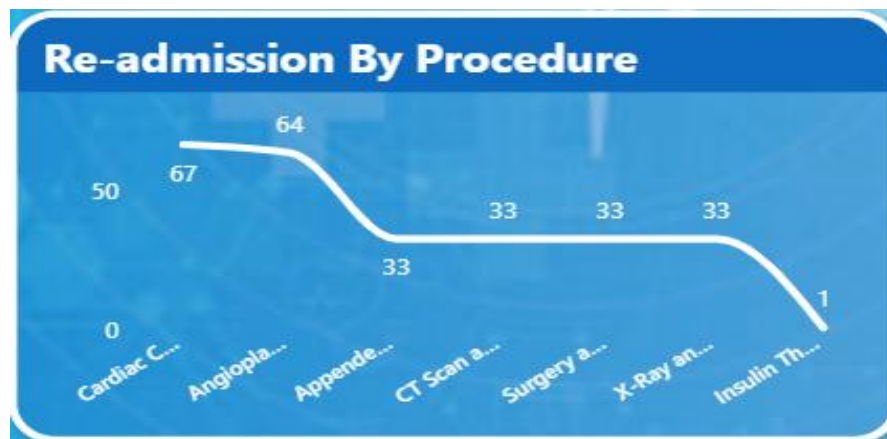
- **Satisfaction by Condition**

This chart shows average satisfaction levels for each condition. It helps understand patient experience across treatments. Useful for service improvement.



- **Readmission by Procedure**

This chart shows procedures associated with higher readmissions. It helps identify risky treatments. Useful for improving medical processes.



Interactivity

- Slicers used:

- Gender

Gender
☐ Female
☐ Male

- Condition

Condition
☐ Allergic Reaction
☐ Appendicitis
☐ Cancer
☐ Childbirth
☐ Diabetes
☐ Fractured Arm
☐ Fractured Leg
☐ Heart Attack
☐ Heart Disease
☐ Hypertension
☐ Kidney Stones
☐ Osteoarthritis
☐ Prostate Cancer
☐ Respiratory Infection
☐ Stroke

- Procedure

Procedure
☐ Angioplasty
☐ Antibiotics and Rest
☐ Appendectomy
☐ Cardiac Catheterization
☐ Cast and Physical Therapy
☐ CT Scan and Medication
☐ Delivery and Postnatal Care
☐ Epinephrine Injection
☐ Insulin Therapy
☐ Lithotripsy
☐ Medication and Counseling
☐ Physical Therapy and Pain Management
☐ Radiation Therapy
☐ Surgery and Chemotherapy
☐ X-Ray and Splint

- Navigation buttons used to switch between pages
- Dynamic filtering applied across visuals

9. Key Insights / Findings

- Certain conditions generate higher treatment costs
- Increase in length of stay leads to higher cost
- Some conditions have higher readmission rates
- Majority of patients recover successfully
- Patient satisfaction varies based on treatment type

10. Business Recommendations

- Focus on reducing readmission rates
- Optimize treatment cost for high-expense conditions
- Improve patient care to increase satisfaction
- Reduce hospital stay duration where possible
- Monitor high-risk procedures closely

11. Challenges Faced

- Handling duplicate values in procedures
- Designing proper KPI measures
- Choosing correct visualizations
- Maintaining clean dashboard layout
- Understanding DAX calculations

12. Conclusion

This Power BI dashboard successfully analyzes hospital data and provides meaningful insights into patient care, financial performance, and risk factors. The interactive dashboard helps stakeholders make better decisions and improve hospital efficiency.

13. Future Improvements

- Add real-time data integration
- Include predictive analytics
- Add more filters (Age group, Outcome)
- Improve UI design further
- Deploy dashboard to Power BI Service

14. References

- Power BI Documentation
- Data Analytics learning resources

Thank You